

ASSIGNMENT NO.4

1. PROBLEM STATEMENT: Write a C Program to calculate the salary of an employee given his basic pay (taken as input from the user). Calculate gross salary of employee. Let HRA be 10 % of basic pay and TA be 5% of basic pay. Let employees pay professional tax as 2% of total salary. Calculate net salary payable after deductions.

OBJECTIVE:

1. To understand the importance of flowchart for any programming model.
2. To learn simple flowchart symbols and arrows to define relationships.
3. To understand and develop visual representations of flow of data.

THEORY:

1. Define Algorithm.
2. What are the different characteristics of an algorithm.

PLATFORM: Linux

CODE:

```
#include <stdio.h>

int main() {
    float basic_pay, hra, ta, gross_salary, professional_tax, net_salary;

    // Input basic pay from user
    printf("Enter the basic pay of the employee: ");
    scanf("%f", &basic_pay);

    // Calculate HRA (10% of basic pay)
    hra = 0.10 * basic_pay;

    // Calculate TA (5% of basic pay)
    ta = 0.05 * basic_pay;

    // Calculate gross salary (basic + HRA + TA)
    gross_salary = basic_pay + hra + ta;

    // Calculate professional tax (2% of gross salary)
    professional_tax = 0.02 * gross_salary;

    // Calculate net salary (gross salary - professional tax)
    net_salary = gross_salary - professional_tax;

    // Display all calculations
    printf("\n=== Salary Breakdown ===\n");
    printf("Basic Pay: Rs. %.2f\n", basic_pay);
    printf("HRA (10%): Rs. %.2f\n", hra);
    printf("TA (5%): Rs. %.2f\n", ta);
    printf("Gross Salary: Rs. %.2f\n", gross_salary);
    printf("Professional Tax (2%): Rs. %.2f\n", professional_tax);
    printf("Net Salary: Rs. %.2f\n", net_salary);

    return 0;
}
```

OUTPUT:

```
Enter the basic pay of the employee: 1000000

=== Salary Breakdown ===
Basic Pay: Rs. 1000000.00
HRA (10%): Rs. 100000.00
TA (5%): Rs. 50000.00
Gross Salary: Rs. 1150000.00
Professional Tax (2%): Rs. 23000.00
Net Salary: Rs. 1127000.00
```

CONCLUSION: Thus, we have learned to write algorithms and to draw its representation in terms of flowchart.